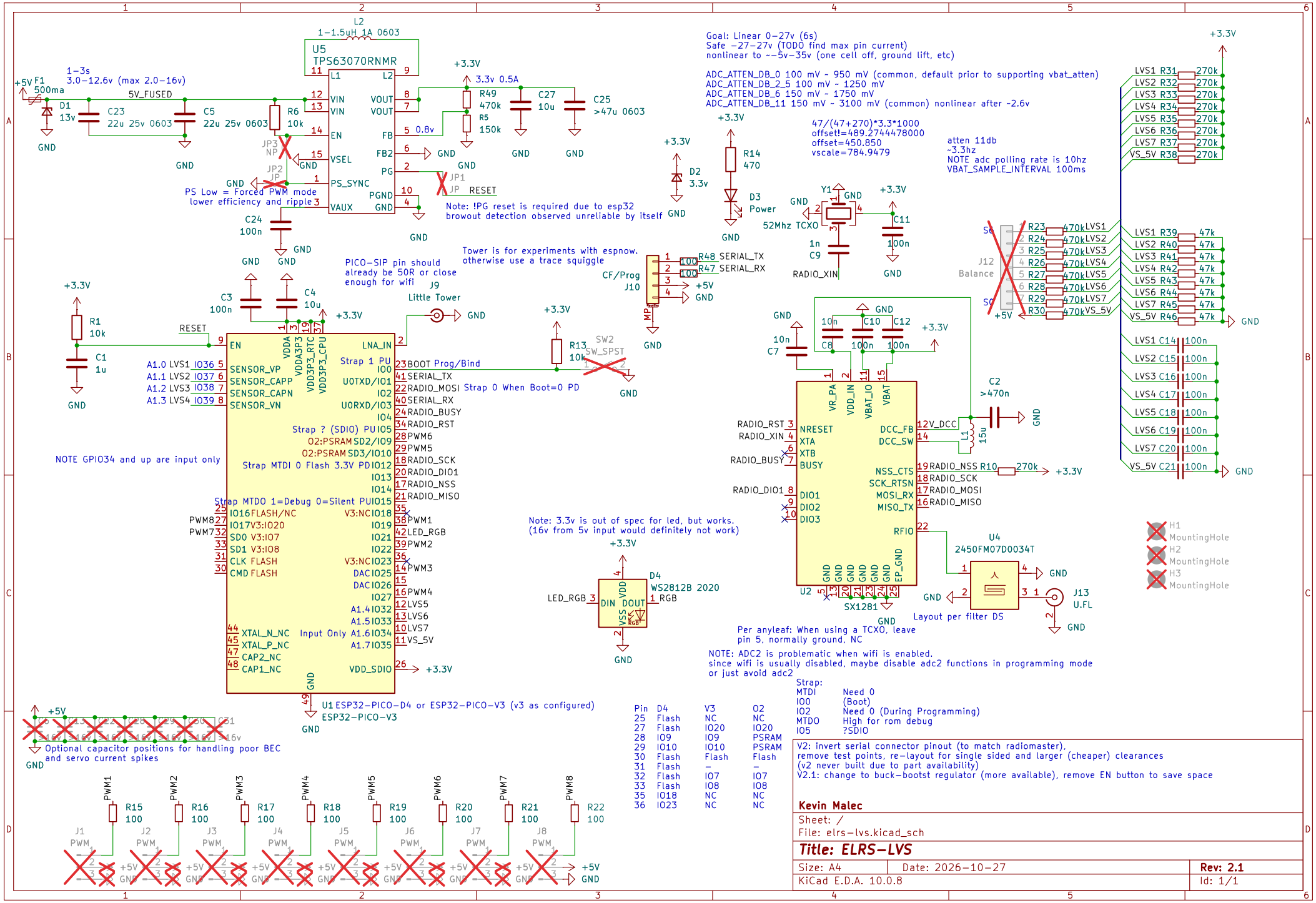
TPSM82901SISR possible alt LMZ21701 expensive but small
TPSM336{20,10,06} slightly larger but more available, TPSM33620S3QRDNRQ1 https://jlcpcb.com/parts
TPS63070 buck/boost highly available but large



Pin	D4	V3	O2
25	Flash	NC	NC
27	Flash	IO20	IO20
28	IO9	IO9	PSRAM
29	IO10	IO10	PSRAM
30	Flash	Flash	Flash
31	Flash	–	–
32	Flash	IO7	IO7
33	Flash	IO8	IO8
35	IO18	NC	NC
36	IO23	NC	NC

Strap:

MTDI	Need 0
IO0	(Boot)
IO2	Need 0 (During Programming)
MTDO	High for rom debug
IO5	?SDIO

V2: invert serial connector pinout (to match radiomaster),
remove test points, re-layout for single sided and larger (cheaper) clearances
(v2 never built due to part availability)
V2.1: change to buck-bootst regulator (more available), remove EN button to save space

Kevin Malec

Sheet: /
File: elrs-lvs.kicad_sch

Title: ELRS-LVS

Size: A4	Date: 2026-10-27	**Rev: 2.1**
KiCad E.D.A. 10.0.8		Id: 1/1